Project Topic Suggestions

The most successful projects combine the tools of Bayesian filtering with an application domain you are already interested, e.g., through your existing research or intended future research. In your project, I will expect you to mathematically formalize your problem, propose an application of existing Bayesian filtering techniques or develop a new problem-specific algorithm, and perform simulation studies in Matlab or another sim environment showing the method in action. Below are some project topics inspired from previous years' projects, mixed with some suggestions from the teaching staff.

- 3D Gaussian Splatting (3DGS) for vision based mapping, localization, SLAM, or object pose estimation.
- Distributed relative localization for a group of robots: a team of robots take noisy relative measurements and communicate over a network to estimation the relative positions of the whole group.
- Distributed multi-target tracking for a group of robots: a team of robots take noisy measurements of a group of dynamic targets, and communicate over a network to estimate the positions of the targets.
- Active mapping: control a robot to drive through an environment to build a high-quality map.
- Distributed SLAM: a team of robots move around an environment and jointly estimate a map and their trajectories in the map, communicating over a network.
- Learning latent dynamical models: Learn a state space model given observations of inputs and outputs. Often times the state is uninterpretable and is called a latent state. One example could be to learn a dynamics model of a pendulum from simulated images (pixel arrays) of the pendulum.
- Gaussian process system ID: Learn a model of a dynamical system using Gaussian Process regression.
- Satellite state estimation with star trackers and IMU: Use nonlinear filter to find the 6 DOF pose of an orbiting satellite with onboard IMU and star tracker.
- Intention inference: Use Bayesian methods to infer the intended motion and/or goal point of an observed target.
- Nonlinear filter of your choice applied to e.g., UAVs, cars, ground robots, satellites, robotic surgical instruments, micro-scale systems, any other domain of interest.